

UNIT 4

UNDERSTANDING COMPUTER FORENSICS: Introduction, Digital Forensics Science, The Need for Computer Forensics, Cyber forensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Network Forensics, Approaching a Computer Forensics Investigation. Forensics and Social Networking Sites: The Security/Privacy Threats, Challenges in Computer Forensics.

1 INTRODUCTION TO COMPUTER FORENSICS

1.1 What is Computer Forensics?

Computer Forensics is the process of:

Identifying, collecting, preserving, analyzing, and presenting digital evidence in a legal manner.

Simple meaning: When a cybercrime happens, experts investigate computers and digital devices to find proof.

1.2 What is Digital Evidence?

Digital evidence = Any information stored or transmitted in digital form that can be used in court.

Examples:

- Emails
- Chat messages
- Hard disk data
- Browser history
- Server logs
- CCTV digital recordings

Engineering View:

Digital forensics = Reverse engineering of crime.

Just like debugging a crashed program, forensics experts “debug” a crime.

2 DIGITAL FORENSICS SCIENCE

Digital Forensics is a branch of forensic science.

It applies:

- Computer science
- Law
- Investigation techniques
- Data recovery methods

2.1 Branches of Digital Forensics

1. Computer Forensics
2. Network Forensics
3. Mobile Forensics
4. Cloud Forensics
5. Database Forensics

Example:

If ransomware attack happens:

- Computer forensics → analyze infected system
- Network forensics → check traffic logs
- Mobile forensics → check communication
- Cloud forensics → check cloud storage

3 THE NEED FOR COMPUTER FORENSICS

Why do we need it?

1. To investigate cybercrime
2. To recover deleted data
3. To identify attacker
4. To present legal proof
5. To prevent future attacks

Real Example:

If company data leaked, forensics helps answer:

- Who accessed it?
- When?

- From which IP?
- Was it internal employee?

Without forensics: No proof → No conviction.

4 CYBER FORENSICS AND DIGITAL EVIDENCE

Cyber forensics deals specifically with crimes involving networks and internet.

4.1 Types of Digital Evidence

1. Volatile Data (Temporary)
 - a. RAM data
 - b. Running processes
 - c. Network connections
2. Non-Volatile Data (Permanent)
 - a. Hard disk
 - b. SSD
 - c. USB drive

Important: Volatile data disappears when power off.

4.2 Properties of Digital Evidence

1. Easily altered
2. Easily copied
3. Must maintain integrity
4. Must be legally admissible

Integrity maintained using:

- Hash functions (SHA-256)
- Write blockers
- Disk imaging

5 FORENSICS ANALYSIS OF E-MAIL

Email is common evidence in cybercrime.

5.1 What Can Be Analyzed?

1. Email header
2. IP address
3. Time stamp
4. Sender route
5. Attachments

5.2 Email Header Analysis

Email header contains:

- Source IP
- Mail server details
- Time of sending

Used to trace: Where email originated.

Example:

If phishing email received, forensics checks:

- Was it spoofed?
- Real server?
- Country of origin?

DIGITAL FORENSICS LIFE CYCLE

Very important for exam.

Step 1: Identification

Recognize that crime happened.

Step 2: Preservation

Secure evidence. Do not modify system.

Use:

- Write blockers
- Disk imaging

Step 3: Collection

Copy data safely.

Step 4: Examination

Extract relevant data.

Step 5: Analysis

Interpret evidence.

Step 6: Presentation

Submit report in court.

Diagram to draw: Flow diagram of above 6 steps.

CHAIN OF CUSTODY CONCEPT

Very important legal concept.

Chain of Custody = Record showing:

- Who collected evidence
- When
- Where
- Who handled it

Why Important?

If evidence tampered → case dismissed.

Example:

If police seize laptop:

- Date recorded
- Officer name recorded
- Storage location recorded

This record ensures authenticity.

NETWORK FORENSICS

Network forensics investigates network traffic.

What Is Analyzed?

1. Packet capture (PCAP files)
2. Firewall logs
3. Server logs
4. IDS alerts

Purpose:

- Detect intrusion
- Trace attacker IP
- Analyze DDoS attack

Engineering Example:

Using packet analysis tools to detect suspicious traffic.

APPROACHING A COMPUTER FORENSICS INVESTIGATION

Investigation must be systematic.

Steps:

1. Obtain legal authorization
2. Secure crime scene
3. Identify devices
4. Create forensic image
5. Analyze evidence
6. Document everything

Important Rule: Never work on original evidence. Always use cloned copy.

10 FORENSICS AND SOCIAL NETWORKING SITES

Social media is major source of digital evidence.

What Can Be Collected?

1. Chat history
2. Photos
3. Location data
4. IP logs
5. Friend connections

Used in:

- Cyber stalking cases
- Harassment cases
- Identity theft cases

1 1 SECURITY / PRIVACY THREATS IN SOCIAL MEDIA

1. Fake profiles
2. Data scraping
3. Identity theft
4. Cyber bullying
5. Phishing links

Example:

Attacker creates fake Instagram profile and collects personal info.

1 2 CHALLENGES IN COMPUTER FORENSICS

12.1 Encryption

If data encrypted, hard to access without key.

12.2 Cloud Storage

Data stored in different countries. Legal jurisdiction problem.

12.3 Large Data Volume

Huge logs → time-consuming analysis.

12.4 Anti-Forensics Techniques

Attackers:

- Delete logs
- Encrypt drives
- Use VPN
- Use proxy chains

12.5 Legal Issues

Privacy laws Cross-border investigation issues

1 Computer Forensics vs Digital Forensics

Basis	Computer Forensics	Digital Forensics
Scope	Focus on computers	Broader (all digital devices)
Devices Covered	PC, Laptop, Hard Disk	Mobile, Network, Cloud, IoT
Area	Subset	Parent field
Example	Analyzing hard disk	Analyzing WhatsApp + Server logs
Use Case	Desktop crime investigation	Cybercrime investigation

2 Volatile Data vs Non-Volatile Data

Basis	Volatile Data	Non-Volatile Data
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Definition	Temporary data	Permanent stored data
Storage	RAM	Hard disk, SSD
Data Loss	Lost when power off	Remains after shutdown
Example	Running processes	Files, documents
Forensic Priority	Must collect first	Can collect later

Important: Volatile evidence is time-sensitive.

3 Identification vs Preservation (Forensic Life Cycle)

Basis	Identification	Preservation
Meaning	Detect crime and evidence	Protect evidence from alteration
Stage	First	Second
Action	Recognize devices	Isolate system
Example	Identify suspect laptop	Use write blocker

4 Collection vs Examination

Basis	Collection	Examination
Meaning	Acquire digital evidence	Extract relevant information
Process	Create forensic image	Recover deleted files
Focus	Data acquisition	Data filtering
Tools Used	Imaging tools	Analysis tools

5 Examination vs Analysis

Basis	Examination	Analysis
Purpose	Extract data	Interpret meaning
Activity	Recover files	Identify suspect behavior
Output	Raw data	Logical conclusion
Example	Extract browser history	Determine timeline

6 Chain of Custody vs Digital Evidence

Basis	Chain of Custody	Digital Evidence
Meaning	Record of handling evidence	Actual digital data
Purpose	Maintain authenticity	Prove crime
Legal Importance	Ensures admissibility	Used in court
Example	Evidence log sheet	Hard disk image

7 Network Forensics vs Computer Forensics

Basis	Network Forensics	Computer Forensics
Focus	Network traffic	Computer system
Data Source	Logs, packets	Hard disk, RAM
Purpose	Trace attack path	Recover files
Example	Analyze DDoS logs	Analyze infected PC

8 Forensics vs Incident Response

Basis	Forensics	Incident Response
Goal	Collect evidence	Stop attack immediately
Time	After attack	During attack
Focus	Legal proof	Damage control
Example	Analyze logs	Disconnect server

9 Encryption vs Anti-Forensics

Basis	Encryption	Anti-Forensics
Purpose	Protect data	Hide criminal activity
Legal Use	Yes	No
Example	Encrypt phone data	Delete logs intentionally
Impact	Security	Investigation difficulty

10 Write Blocker vs Disk Imaging

Basis	Write Blocker	Disk Imaging
Function	Prevent data modification	Create exact copy
Usage	During evidence access	During evidence collection
Purpose	Protect integrity	Preserve original
Example	USB write blocker	Bit-by-bit clone

1 1 Email Header vs Email Body

Basis	Email Header	Email Body
Contains	Routing information	Message content
Use in Forensics	Trace IP	Analyze communication
Visible to User	Mostly hidden	Visible
Example	Server path	Text message

1 2 Social Media Evidence vs Traditional Evidence

Basis	Social Media Evidence	Traditional Evidence
Nature	Digital	Physical
Easily Altered	Yes	Harder
Storage	Cloud servers	Physical location
Example	Chat logs	Paper documents

1 3 Digital Forensics vs Traditional Forensics

Basis	Digital Forensics	Traditional Forensics
Evidence Type	Digital data	Physical objects
Tools	Software tools	Physical lab tools
Complexity	High (encryption)	Physical analysis
Example	Hard disk analysis	Fingerprint analysis

